

GROWRIDER · REQUIREMENTS

GrowRider requirements

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DOCUMENT CONTROL

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Project TRL	TRL 3 Proof of concept		
Repository	github.com/BoujeeEnjinia1701/growrider		
Source	docs/03-requirements.md @ uncommitted		

REVISION HISTORY

Version	Date	Author	Change
0.3	2026-09-25	Amish Chadha	Record decisions of GRR-DDR-001 (20 in wheels, brakes, \$120 production target); add TRL 3 status from GRR-CAL-001
0.2	2026-09-24	Amish Chadha	First measurable requirements for TRL 2
0.1	2026-09-24	Amish Chadha	Initial scaffold

GrowRider requirements

These are the requirements for the concept. They were checked by calculation at TRL 3 in GRR-CAL-001: 5 are met on paper, 1 is not met (R5, mass), 4 are at risk and 2 cannot be verified at TRL 3. No target was relaxed or redefined by the TRL 2 review decisions (GRR-DDR-001). All targets must still be revised from co-design findings before the design is frozen (see GRR-PRB-001).

ID	Requirement	Target	Verification (TRL 3 or later)	TRL 3 status (GRR-CAL-001)
R1	Fit the intended riders	Rider height 1.10 to 1.65 m (ages about 6 to 14); standover height 470 mm or less at the stepping point	Fit calculation from anthropometric data; later fit trials with children	Met: standover 455 mm
R2	Adjustable saddle height	436 to 654 mm from bottom bracket center to saddle top (218 mm range), with at least 100 mm insertion at every sliding joint across the range	Geometry calculation; massing model	Met: 400 to 670 mm, 100 mm insertion
R3	Adjustable reach and handlebar height	Handlebar height adjustable by 100 mm or more; horizontal saddle-to-grip reach adjustable by 100 mm or more	Geometry calculation	Met: bar 113 mm, reach 141 mm
R4	One wheel size across the whole fit range	Single wheel size for all riders: 20 in (ISO 406), decided 2026-09-25 (GRR-DDR-001, D1)	Design review; fit calculation	Met: 20 in
R5	Light enough for the smallest rider to handle	Complete bike with rack, fenders and kickstand 13 kg or less	Mass estimate, then weighing	Not met: 15.3 kg
R6	Small, clearly rated rear rack	Rated 10 kg, rating marked on the rack; platform no larger than a school bag (about 300 x 140 mm); no passenger footrests	Design review; later static load test	Met by design: 300 x 120 mm
R7	Two independent brakes a child can operate	Rear coaster brake plus front rim brake (decided 2026-09-25, GRR-DDR-001, D2); each brake alone 0.2 g or more and both together 0.35 g or more on a dry, level surface; hand lever reach adjustable for small hands	First-order braking calculation; later tests to ISO 8098 or ISO 4210 as applicable	At risk: small margins for the smallest rider; wet steel rim poor
R8	Puncture resistance	No loss of use from thorn punctures; tire life 2,000 km or more (about one school year of 10 km per day)	Supplier data; later field trial	Not verifiable at TRL 3 (tire life)
R9	Serviceable with basic tools	All fit adjustments and routine service with 13 mm and 15 mm spanners and a screwdriver; a fit change takes 10 minutes or less for a parent or teacher	Design review; later timed trial	At risk: 10 min; headset and BB need two more tools
R10	Parts commonality with regional adult roadsters	Every wear part except tires and rims interchangeable with regional roadster parts (chain, sprocket, coaster hub internals, bottom bracket, headset, pedals, seat post, handlebar, brake blocks)	BOM review against a regional parts list from the partner	Not verifiable at TRL 3 (no regional parts list)
R11	Low cost	Prototype parts \$250 or less (project budget); production cost target \$120 or less per bike at volume (decided 2026-09-25, GRR-DDR-001, D6)	Priced BOM; production estimate with a regional assembler	At risk: \$243 bike, \$255 with helmet
R12	Lasts across siblings	10-year service life for at least three successive riders; frame and fork sized for a 60 kg rider plus		

ID	Requirement	Target	Verification (TRL 3 or later)	TRL 3 status (GRR-CAL-001)
		a 10 kg rack load; sliding joints that do not seize in dust and rain	Fatigue calculation at TRL 3; later frame fatigue tests	At risk: 3 of 6 sections above the fatigue screen

■ Assumptions

- Inseam is about 0.45 times standing height, and a comfortable saddle height (bottom bracket center to saddle top) is about 0.88 times inseam. Both are common bicycle fitting rules of thumb for adults and older children and must be checked against measurements of the intended users.
- Horizontal saddle-to-grip reach for an upright riding position scales with torso and arm length, estimated at about 0.2 times height (220 mm at 1.10 m, 330 mm at 1.65 m). This is a placeholder until fit measurements are taken.
- A typical school trip is 5 km each way on dirt or gravel, ridden at 10 to 12 km/h.
- A 1.65 m rider is assumed to weigh up to 60 kg, including a margin for older teenagers.
- Saddle height (R2) is measured from the BB center to the saddle top along the seat axis. GRR-CAL-001 also checks a second fitting rule that accounts for the 140 mm crank (saddle top to pedal at 1.09 times inseam), which asks for 400 to 669 mm; the design covers both.
- Applicable standard: classified by its maximum saddle height above the ground (about 890 mm), GrowRider falls under ISO 4210-2 as a city and trekking bicycle; ISO 8098 covers maximum saddle heights below 635 mm (GRR-CAL-001, section M).